

Frontend User Interface Requirement Specification (FURS)

1. Introduction

1.1 Purpose

This document specifies the user interface requirements for the Ride Sharing Dispatch System frontend application.

1.2 Scope

Explain:

- Rider interface: The main interface persons seeking a ride will use
 - Driver interface: The main interface persons willing to drive will use
 - Authentication UI: Login and Registration pages
-

2. Overall Description

2.1 User Roles

Rider

- Can register
- Can log in
- Can create trip
- Can cancel trip
- Can view history

Driver

- Can register
 - Can log in
 - Can toggle availability
 - Can update location
 - Can accept trip
-

3. UI Functional Requirements

3.1 Authentication Interface

- The system shall provide a login page.
 - The system shall validate email and password fields.
 - The system shall redirect users based on role after login.
 - The system shall store JWT securely in browser storage.
-

3.2 Rider Dashboard Requirements

- The system shall display a map centered on the rider's location.
- The system shall allow selection of pickup and destination.
- The system shall allow vehicle type selection.
- The system shall prevent trip creation if an active trip exists.
- The system shall display current trip status.
- The system shall display trip history.

3.3 Driver Dashboard Requirements

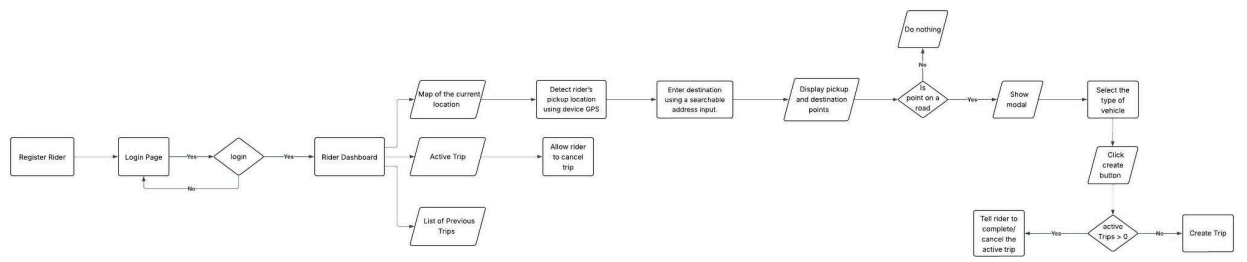
Example:

- The system shall display driver availability status.
- The system shall allow toggling between online and offline.
- The system shall prevent going offline during active trip.
- The system shall allow driver to update current location.
- The system shall show assigned trip details.

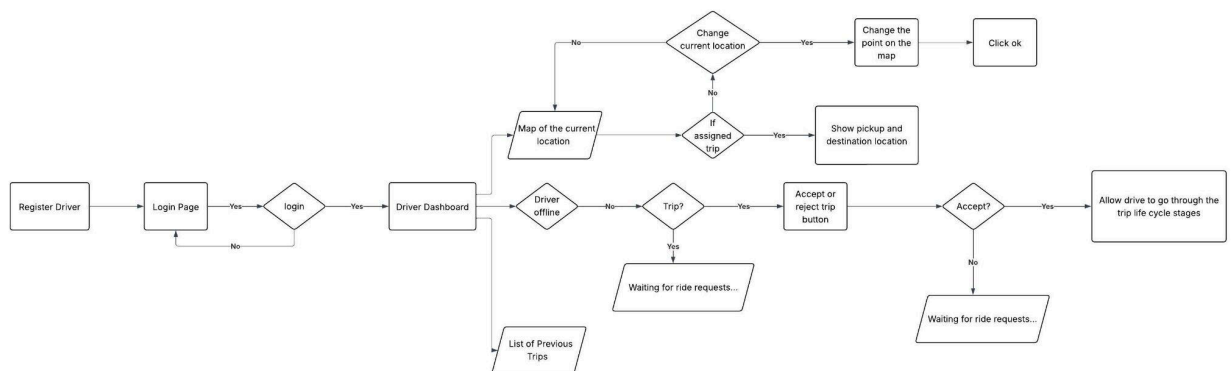
Right now this section is probably missing or weak.

4. User Flow Diagrams

- Rider Flow



- Driver flow



5 Error Handling & Exception Management

5.1 General Error Handling Requirements

The system shall:

- Display user-friendly error messages.
 - Prevent system crashes or blank screens.
 - Allow the user to retry failed actions where applicable.
 - Log technical errors internally for debugging purposes.
-

5.2 API Failure Handling

If an API request fails due to:

- Server error (5xx),
- Bad request (4xx),
- Timeout,
- Network interruption,

The system shall:

1. Display a non-technical message such as:

“We couldn’t process your request. Please try again.”

2. Keep the user on the current page.
3. Allow the user to retry the action.

The system shall not display raw server error messages.

5.3 Authentication & Session Expiry

If a request returns a 401 Unauthorized response:

1. The system shall automatically clear the stored authentication token.
 2. The system shall redirect the user to the Login page and display “Your session has expired. Please log in again.”
-

5.4 Network Connectivity Failure

If the application detects loss of internet connectivity:

1. The system shall display a notification banner indicating that the user is offline.
 2. The system shall temporarily disable trip creation and status update actions.
 3. The system shall automatically restore functionality when connectivity is re-established.
-

5.5 Map Component Failure

If the map service fails to load due to external API failure or network issues:

1. The system shall display a fallback interface allowing manual address entry and display “Map failed to load. You may enter addresses manually.”

The system shall remain usable without map functionality.

5.6 No Available Drivers

If no drivers match the rider’s request:

1. The system shall notify the rider:
“No drivers available nearby. Please try again shortly.”
 2. The rider shall be allowed to retry the request.
 3. The trip shall remain in “Requested” status.
-

5.7 Active Trip Restriction

If a rider attempts to create a new trip while an active trip exists:

1. The system shall prevent the creation of a new trip and display “Please complete or cancel your active trip before creating a new one.”
-

5.8 Driver Availability Constraints

If a driver attempts to switch to offline mode during an active trip:

1. The system shall prevent the status change and display “You cannot go offline while a trip is in progress.”